

Quantization outside of Efficiency

How does it differ in real-life scenarios?

Outline

- Quantization is essential for LLM inference
- Efficiency seems to be well studied in benchmarks
- Compare (Novel) quantization methods (e.g. native quantization) with current methods
- What about bias, interpretability and trustworthiness in real-life scenarios?

Goals of the project

- Research Question #1: How do quantization methods impact metrics beyond perplexity and accuracy e.g. bias, trustworthiness, interpretability
- Research Question #2: Does the gap on these metrics between standardized benchmarks and non-standardized benchmark performance (e.g. counterfactuals) increase with quantization methods?
 - Intuition: Real-world scenario vs benchmark gap

Division of Responsibility

- Yingte: Implementation of Quantization Methods
- Ramneek and Hans: Evaluation/Conceptual Side

Project Outline

- Two parts
 - First: Implementing
 - Second: Evaluating
- Implementation Native Quantization and the combination of quantization and pruning
- Models Open Source Candidates:
 - Qwen
 - Deepseek
 - Llama
 - Mistral
- Size constraint: Max 4B
- Quantization
 - FP16
 - Current Method
 - Novel Method

Project Timeline I

- First Week:
 - Literature Research
 - Report skeleton (e.g. background)
 - Project Setup and Code Skeleton
 - Smoke Test Pipeline (e.g. first experiments)
- Second Week:
 - First Round of Iteration
 - Implementation
 - And Analysing the Results

Project Timeline II

- Third Week:
 - Further iterate on the finding on the second week
 - Refine the code base
 - Finalize results
- Final Week:
 - Write up the report
 - Finish presentation